

Notes: Guerrieri and Lorenzoni (QJE, 2017)

Credit Crises, Precautionary Savings, and the Liquidity Trap

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1 Big picture

- **Question.** How does an economy adjust when households suddenly face tighter borrowing conditions? What happens to interest rates, consumption, employment, and output during the deleveraging process?
- **Setting.** The paper studies a heterogeneous-agent incomplete-markets economy of the Bewley-Aiyagari type in which households borrow and lend to smooth idiosyncratic labor-income risk.
- **Core mechanism.** A credit crunch pushes **borrowers** to reduce debt and also induces **lenders** to save more for precautionary reasons. Both forces raise desired net lending, so the equilibrium real interest rate must fall.
- **Main theoretical result.** The natural interest rate falls sharply on impact and then gradually rises toward a lower steady state. The initial fall *overshoots* because the economy starts with too many households near the lower tail of the wealth distribution, all trying to save at once.
- **Main quantitative result.** In the baseline flexible-price economy, a tightening of the borrowing limit that lowers the long-run debt-to-GDP ratio by 10 percentage points causes output to fall by about **1.1%** on impact.
- **Nominal rigidity result.** With fixed nominal wages and a zero lower bound on nominal rates, the central bank cannot replicate the flexible-price natural-rate path, so the contraction is larger: output falls by about **1.7%**.
- **Mechanism on the demand side.** The partial-equilibrium drop in consumption from tighter credit is large (about 4%), but is strongly offset by the endogenous fall in interest rates.
- **Mechanism on the supply side.** Low-wealth households work more to deleverage, but the extra hours come disproportionately from low-productivity workers, so average labor productivity falls. In the baseline, aggregate hours rise while output falls.
- **Durables extension.** When households hold both liquid bonds and illiquid durables used as collateral, a credit shock can simultaneously increase demand for liquid assets and decrease demand for illiquid assets. A transitory borrowing-spread shock then generates a much larger contraction, concentrated in durable expenditure.
- **Why this paper matters.** The paper gives a heterogeneous-agent foundation for why credit crises can produce very low natural interest rates and liquidity traps, instead of imposing ad hoc preference shocks as in representative-agent models.

2 Data and measurement (key objects)

This is a **quantitative theory paper**, not a reduced-form empirical paper. There is no estimation sample in the usual firm- or household-level sense. The “data” are the calibration targets and the model’s state variables.

2.1 Observed household-state space in the model

- **Households.** A continuum of households indexed by i .

- **Idiosyncratic states.** Each household has bond holdings b_{it} and labor-productivity state θ_{it} .
- **Unemployment state.** The lowest productivity state satisfies $\theta_1 = 0$ and is interpreted as unemployment.
- **Aggregate state.** The key aggregate state variable is the full joint distribution

$$\Gamma_t(b, \theta),$$

not just a small set of moments.

2.2 Calibration targets and measured moments

The baseline calibration uses U.S. moments to discipline preferences, income risk, household balance sheets, and labor-market transitions.

- **Interest rate target.** β is chosen so that the initial steady-state annual interest rate is **2.5%**.
- **Risk aversion.** The baseline coefficient of relative risk aversion is $\gamma = 4$.
- **Labor supply elasticity.** η is chosen to match an average Frisch elasticity of 1.
- **Hours worked.** ψ is chosen so that employed workers supply hours equal to 40% of the time endowment, following Nekarda and Ramey.
- **Wage process.** The positive productivity states follow an AR(1) process in logs, calibrated to match the persistence and variance of yearly wages in Floden and Linde.
- **Employment transitions.** Quarterly separation and job-finding rates are calibrated from Shimer: $\pi^{e,u} = 0.057$ and $\pi^{u,e} = 0.882$.
- **Unemployment insurance.** The benefit ν is set to 40% of average labor income.
- **Liquid assets.** The aggregate bond supply B is chosen to match a 2006 U.S. liquid-assets-to-GDP ratio of about **1.78**.
- **Consumer debt.** The borrowing limit ϕ is chosen to match consumer credit equal to about **18% of GDP** in 2006.

2.3 Unobservables and timing

- **Unobservable risk.** Future labor income risk is idiosyncratic and uninsured.
- **Timing.** The paper studies transitional dynamics after a one-time, unexpected aggregate tightening in borrowing conditions.
- **Shock path.** In the baseline experiment, the borrowing limit falls gradually over six quarters from the old steady state to a tighter permanent level.
- **No aggregate capital in baseline.** The paper deliberately abstracts from productive capital and focuses on the household side of the credit crisis.

2.4 Quantitative exercises and applications

- baseline permanent tightening of the borrowing limit,
- economy with nominal wage rigidity and the zero lower bound,
- alternative calibrations that change liquid wealth targets, labor supply curvature, and risk aversion,
- debt-financed fiscal transfers and debt deflation,
- an extension with durable goods, collateralized borrowing, and a borrowing-lending spread.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline environment

Households maximize

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t U(c_{it}, n_{it}) \right],$$

with separable isoelastic utility

$$U(c, n) = \frac{c^{1-\gamma}}{1-\gamma} + \psi \frac{(1-n)^{1-\eta}}{1-\eta}.$$

Each household produces consumption goods with linear technology

$$y_{it} = \theta_{it} n_{it},$$

where θ_{it} is an idiosyncratic labor-productivity shock.

- **Interpretation.** The production side is deliberately simple. All the action comes from heterogeneity in household wealth and earnings risk, not from firm dynamics or capital adjustment.
- **Why this matters.** The paper isolates the household-balance-sheet channel of a credit crunch.

3.2 Budget constraint, borrowing limit, and government

The household budget constraint is

$$q_t b_{it+1} + c_{it} \leq b_{it} + y_{it} - \tilde{\tau}_{it},$$

where b_{it} is bond holdings, q_t is the bond price, and taxes satisfy

$$\tilde{\tau}_{it} = \tau_t \quad \text{if } \theta_{it} > 0, \quad \tilde{\tau}_{it} = \tau_t - \nu_t \quad \text{if } \theta_{it} = 0.$$

The exogenous borrowing constraint is

$$b_{it+1} \geq -\phi. \tag{1}$$

The real interest rate is implicit in the bond price:

$$r_t = \frac{1}{q_t} - 1.$$

Government budget balance is

$$B_t + uv_t = q_t B_{t+1} + \tau_t,$$

where u is the unemployment rate.

- **Interpretation.** The debt limit ϕ is the object hit by the credit crunch. Tighter ϕ means less borrowing capacity.
- **Asset supply.** The outside supply of liquid assets comes from government bonds, interpreted broadly as the net supply of liquid assets available to households.
- **Important simplification.** The baseline model contains no capital and only one liquid asset, so the interest rate is the single price that equilibrates saving and borrowing.

3.3 Household optimality conditions

The Euler inequality is

$$U_c(c_{it}, n_{it}) \geq \beta(1 + r_t) \mathbb{E}_t[U_c(c_{it+1}, n_{it+1})], \quad (2)$$

with equality when the borrowing constraint does not bind.

The labor-supply condition is

$$\theta_{it} U_c(c_{it}, n_{it}) + U_n(c_{it}, n_{it}) \leq 0, \quad (3)$$

with equality if $n_{it} > 0$.

- **Precautionary-savings channel.** A tighter borrowing constraint makes future consumption more sensitive to income shocks, which raises expected marginal utility tomorrow and pushes consumption down today.
- **Microfoundation for a demand shock.** In that sense, a credit crunch in a heterogeneous-agent model behaves like a negative aggregate demand shock, but it is generated endogenously from tighter borrowing capacity.

3.4 Equilibrium and the role of the wealth distribution

Let $\Gamma_t(b, \theta)$ denote the joint distribution of wealth and productivity. An equilibrium is a sequence of interest rates, policies, taxes, and distributions such that

- household policies are optimal,
- the distribution evolves consistently with those policies and the Markov process for θ ,
- the government budget constraint holds,
- the bond market clears:

$$\int b d\Gamma_t(b, \theta) = B.$$

- **Interpretation.** Unlike approximate heterogeneous-agent business-cycle models that reduce the distribution to a few moments, this paper keeps the full wealth distribution as a state variable.
- **Why this matters.** The paper’s central result — the overshooting of the natural interest rate — is driven by how the lower tail of the wealth distribution adjusts over time.

3.5 The credit-crunch experiment

The baseline experiment starts from a steady state with $\phi = 0.959$ and then imposes an unexpected permanent tightening to

$$\phi' = 0.525,$$

chosen so that the long-run debt-to-GDP ratio falls by 10 percentage points.

Starting at $t = 1$, the borrowing limit follows

$$\phi_t = \max\{\phi', \phi - \Delta\phi \cdot t\},$$

with the adjustment spread over six quarters.

- **Interpretation.** The gradual path is a reduced-form way to capture the fact that actual household debt has maturity longer than one quarter, so deleveraging is not fully instantaneous.
- **Steady-state implication.** The tighter borrowing limit shifts aggregate bond demand to the right through both direct deleveraging and higher precautionary saving, lowering the long-run equilibrium interest rate.

3.6 Why the interest rate overshoots

The paper’s first main result is that the interest rate falls sharply on impact and then rises gradually toward a lower steady state.

- In the new steady state, the bond-accumulation policy is more aggressive at low wealth because households want to stay farther away from the borrowing limit.
- At high wealth, the lower long-run interest rate makes households decumulate assets faster.
- This combination makes the new wealth distribution more concentrated than the old one.
- If the interest rate were to jump immediately to its new steady-state level while the old distribution were still in place, aggregate bond demand would be too high because there are “too many” households near the lower tail still trying to save.

Therefore, the equilibrium interest rate has to fall *below* its new steady-state level at first to clear the asset market.

- **Key economic intuition.** Overshooting is not caused by an exogenous taste shock; it comes from the interaction of convex saving behavior with a wealth distribution that initially has too much mass among low-wealth households.

3.7 Output and consumption decomposition

Output falls by about 1.1% on impact in the flexible-price baseline. The paper decomposes consumption into:

- (a) the **partial-equilibrium effect** of the tighter debt limit, keeping the interest rate fixed, and
- (b) the **general-equilibrium effect** from the endogenous fall in interest rates.

A key approximation is

$$\Delta C^{PE} \approx MPC \cdot \Delta \phi,$$

where MPC is the propensity to consume out of a one-time transfer.

Numerically, the baseline calibration gives:

- a partial-equilibrium contraction in consumption of about **3.8%**,
- an offsetting consumption increase of about **2.7 percentage points** from the fall in interest rates.
- **Interpretation.** Tighter credit alone would generate a much larger recession. The endogenous fall in the natural interest rate is a powerful stabilizer.
- **Distributional implication.** Low-wealth households cut consumption the most because they have high MPCs; high-wealth households respond more to lower interest rates and can even increase consumption modestly.

3.8 Employment and composition effects

In partial equilibrium, tighter credit makes indebted households want to work more and consume less. The fall in the interest rate works in the opposite direction by encouraging intertemporal substitution toward less work today.

In the baseline, total hours rise, but output falls because the increase in hours is concentrated among low-productivity, low-wealth households, while high-productivity savers reduce hours more.

- **Interpretation.** Aggregate hours and aggregate output need not move together in this model.
- **Key mechanism.** There is a strong **composition effect**: deleveraging reallocates labor effort toward low-productivity agents, reducing average labor productivity.

3.9 Nominal rigidities and the zero lower bound

To introduce nominal rigidity, the paper assumes fixed nominal wages and therefore a fixed price level. Labor-market rationing is captured by a wedge ω_t in the labor-supply condition:

$$(1 - \omega_t)\theta_{it}U_c(c_{it}, n_{it}) + U_n(c_{it}, n_{it}) \leq 0. \quad (4)$$

The central bank sets the nominal interest rate to replicate the flexible-price allocation whenever possible, subject to

$$r_t \geq 0.$$

- **Interpretation.** The flexible-price allocation requires a strongly negative real rate after the credit crunch. Under fixed wages, the nominal rate equals the real rate, so the zero lower bound may bind.
- **Result.** When the ZLB binds, the economy cannot generate enough interest-rate adjustment, so consumption and output fall more and deleveraging is slower because income is depressed.

3.10 Alternative calibrations

The paper shows that the quantitative strength of the recession depends on how the model is disciplined.

- **Targeting median rather than average liquid wealth** produces lower liquid wealth in the model, larger MPCs, and lower interest elasticity of aggregate consumption. Output falls more sharply: about **1.8%** with flexible prices and about **5%** with nominal rigidity.
- **Lower ψ / different labor-supply curvature** can make aggregate employment fall, not rise, after the credit crunch.
- **Combining lower wealth and lower ψ** can generate a permanently negative natural rate — the paper connects this case to secular-stagnation logic.
- **Higher risk aversion γ** also deepens the downturn because it lowers the intertemporal elasticity of substitution and weakens the stabilizing effect of falling interest rates.

3.11 Fiscal policy and debt deflation

The paper studies debt-financed transfers using a government bond path

$$B_t = \rho_b^t B + (1 - \rho_b^t) B',$$

where the long-run bond supply B' is 20% above its initial level.

Two transfer policies are compared:

- a temporary reduction in the lump-sum tax for everyone,
- a temporary 50% increase in unemployment benefits.
- **Result.** Both policies damp the recession by increasing the supply of safe assets and supporting demand, but transfers targeted to the unemployed are much more powerful.
- **Debt-deflation experiment.** If debt is nominal and the price level unexpectedly falls by 10%, the real debt burden rises and the recession becomes sharper, in line with Fisher’s debt-deflation logic.

3.12 Durable goods, collateral, and credit spreads

In the extension, households choose nondurable consumption c_{it} , durable stock k_{it} , and bond holdings. Durables are illiquid because selling them incurs a proportional cost. Borrowing is collateralized by durables:

$$b_{it+1} \geq -\phi_k k_{it+1}. \tag{5}$$

Borrowers also pay an intermediation spread. If $b_{it+1} < 0$, they receive the net bond price

$$\hat{q}_t = (1 - \chi)q_t,$$

where χ is the intermediation cost.

- **Permanent collateral shock.** A permanent fall in ϕ_k from 0.8 to 0.56 lowers the debt-to-GDP ratio from 54% to 44%. In this case, nondurable spending falls but durable spending rises enough that output actually increases slightly.
- **Transitory spread shock.** A temporary rise in χ affects all borrowers, not just those near the collateral constraint. The result is a much larger downturn concentrated in durables: output falls by about **3.5%** and durable purchases by about **17.2%**.
- **Interpretation.** In a model with liquid and illiquid assets, a credit shock can raise demand for liquid safe assets while reducing demand for illiquid assets such as durables.

4 Identification and quantitative discipline (what makes the results credible)

4.1 There is no reduced-form identification in the usual econometric sense

This is not an instrumental-variables or natural-experiment paper. The paper's credibility comes from the logic of a disciplined quantitative model rather than from statistical identification of a treatment effect.

- The paper does **not** claim to estimate a causal elasticity from quasi-experimental data.
- Instead, it asks whether a standard heterogeneous-agent model with realistic balance-sheet heterogeneity can quantitatively generate low natural rates, output contractions, and liquidity-trap dynamics after a credit crunch.

4.2 What disciplines the model

The model is pinned down by a small set of moments with transparent economic meanings:

- household liquid assets,
 - consumer credit,
 - the wage process,
 - unemployment transitions,
 - the steady-state interest rate,
 - average hours worked.
- **Why this matters.** The paper is not free to generate any recession dynamics it wants. The same calibration must match U.S.-style household balance sheets and labor-market risk first.

4.3 Why the wealth distribution is the key state variable

The central propagation mechanism is the endogenous evolution of the wealth distribution.

- Models with only a representative borrower and lender mute this channel.
- Here, the overshooting of interest rates and the persistence of deleveraging come from how the lower tail of the distribution slowly moves away from the borrowing limit.
- **Core discipline.** If the model were collapsed to a two-type environment, the strongest novel result of the paper — the large initial fall in the natural rate caused by the distributional transition — would disappear or become much weaker.

4.4 Why the robustness exercises matter

The paper is unusually explicit that its quantitative results depend on calibration choices.

- Targeting average versus median liquid wealth changes MPCs and interest-rate elasticities.
- The labor-supply curvature changes whether employment rises or falls.
- Higher risk aversion changes the strength of both precautionary savings and intertemporal substitution.
- **Interpretation.** The paper is credible precisely because it shows where the mechanism is robust and where the quantitative magnitudes are sensitive.

4.5 Relation to the literature

- Relative to representative-agent liquidity-trap models, this paper gives a household-balance-sheet origin for the fall in the natural rate.
- Relative to two-agent deleveraging models, it adds precautionary saving and full wealth-distribution dynamics.
- Relative to the Bewley-Aiyagari literature, it focuses on transitional dynamics after an aggregate tightening in borrowing conditions rather than on steady states or small business-cycle shocks.

5 Tables and figures

5.1 Table I: Baseline calibration

Read:

- The table shows how the model is pinned down by transparent U.S. moments: a 2.5% interest rate, labor-market risk, 40% hours worked, liquid assets, and consumer credit.
- The key balance-sheet targets are a liquid-assets-to-GDP ratio around 1.78 and consumer credit around 18% of GDP.

TABLE I
PARAMETER VALUES

Parameter	Explanation	Value	Target/source
β	Discount factor	0.9711	Interest rate $r = 2.5\%$
γ	Coefficient of relative risk aversion	4	
η	Curvature of utility from leisure	1.5	Average Frisch elasticity = 1
ψ	Coefficient on leisure in utility	12.48	Average hours worked 0.4 of endowment (Nekarda and Ramey 2010)
ρ	Persistence of productivity shock	0.967	Persistence of wage process in Flodén and Lindé (2001)
σ_ε	Variance of productivity shock	0.017	Variance of wage process in Flodén and Lindé (2001)
$\pi_{e,u}$	Transition to unemployment	0.057	Shimer (2005)
$\pi_{u,e}$	Transition to employment	0.882	Shimer (2005)
ν	Unemployment benefit	0.10	40% of average labor income
B	Bond supply	1.6	Liquid assets (flow of funds)
ϕ	Borrowing limit	0.959	Total gross debt (flow of funds)

Note. See the text for details on the targets.

- The chosen borrowing limit $\phi = 0.959$ is large in levels because the model is quarterly and debt is measured relative to average income.
- This table is important because the entire quantitative exercise rests on a calibration that looks like the pre-crisis U.S. household sector.

5.2 Figure I and Figure II: Steady-state policies and bond-market equilibrium

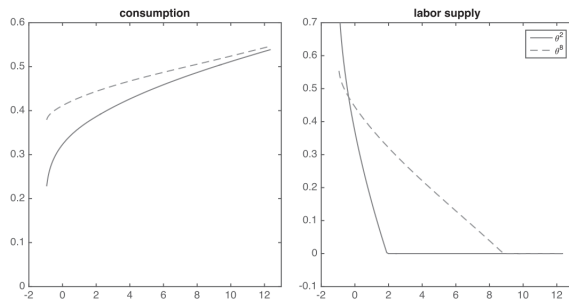


FIGURE I
Optimal Consumption and Labor Supply in Steady State

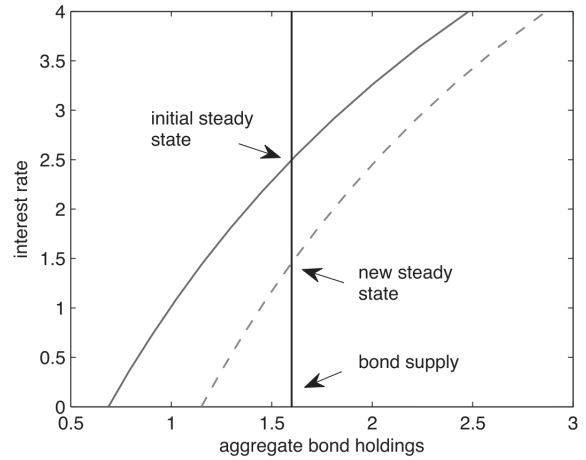


FIGURE II
Bond Market Equilibrium in Steady State
Interest rate is in annual terms.

Read:

- Figure I shows that consumption is concave in wealth and labor supply is strongly decreasing in wealth, especially at low wealth levels. This is where precautionary behavior is strongest.

- The convex labor-supply schedule at low wealth is crucial for the baseline prediction that aggregate hours rise after the credit crunch.
- Figure II shows that tightening the borrowing limit shifts bond demand to the right even in steady state.
- The rightward shift comes from both direct deleveraging and precautionary savings, which is why the long-run interest rate falls.

5.3 Figure III and Figure IV: Interest-rate overshooting and wealth-distribution dynamics

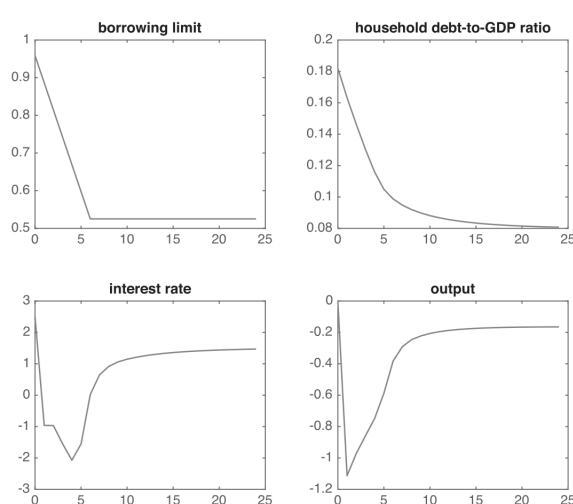


FIGURE III
Interest Rate and Output Responses

Interest rate is in annual terms. Output is in percent deviation from initial steady state.

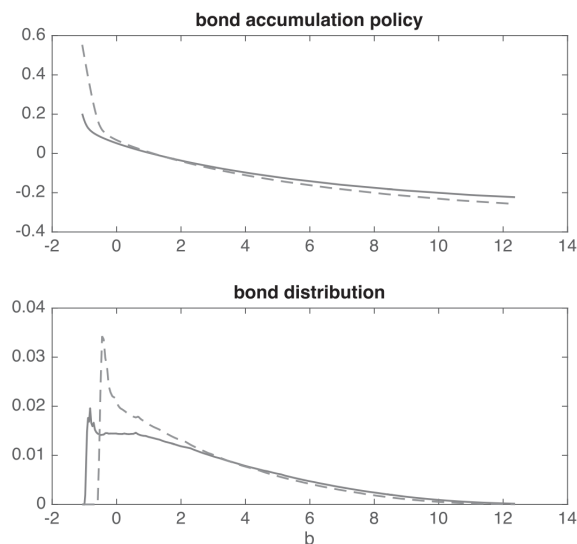


FIGURE IV

Bond Accumulation and Distributions in the Two Steady States
Solid line: initial steady state. Dashed line: new steady state.

Read:

- Figure III contains the central baseline result: after the borrowing limit falls, the interest rate plunges sharply below its new long-run level and output drops by about 1.1%.
- The debt-to-GDP ratio gradually falls as households deleverage.
- Figure IV explains why the interest rate overshoots: with the old wealth distribution still in place, too many households near the lower tail try to save at the new steady-state rate.
- The interest rate must therefore fall more on impact and then gradually rise back as the wealth distribution itself adjusts.

5.4 Figure V, Figure VI, and Figure VII: Consumption decomposition, heterogeneity, and employment

Read:

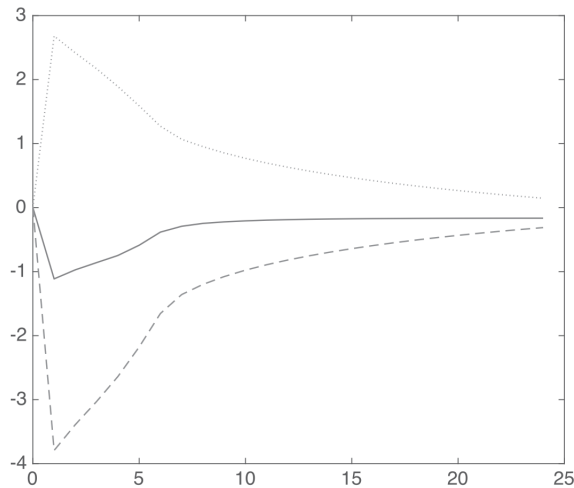


FIGURE V

Consumption Response Decomposition

Percent deviations from initial steady state. Solid line: general equilibrium response. Dashed line: partial equilibrium response to debt limit reduction. Dotted line: response to the equilibrium sequence of interest rate changes.

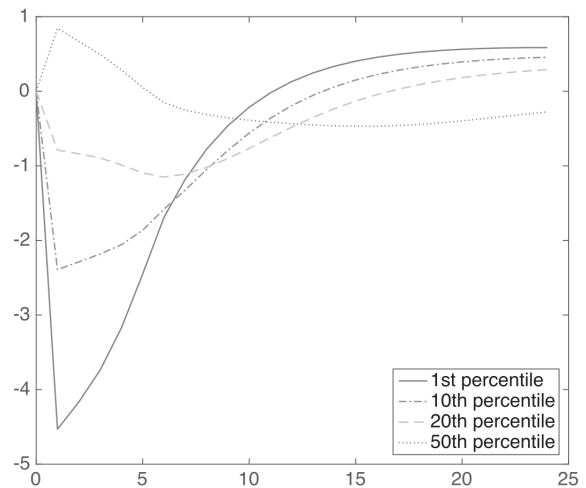


FIGURE VI

Consumption Response by Percentile in Initial Wealth Distribution

Percent deviations from steady-state path conditional on initial wealth being in the reported percentile.

- Figure V shows that the credit shock alone would generate a large consumption drop, but the endogenous fall in interest rates offsets much of it.
- Quantitatively, the partial-equilibrium contraction is about 3.8%, while lower rates add back about 2.7 percentage points.
- Figure VI shows the cross section: low-wealth households cut consumption sharply, while high-wealth households react less or may even raise consumption slightly.
- Figure VII shows that hours rise in the baseline, but output still falls because labor effort shifts toward low-productivity households.

5.5 Figure VIII: Nominal rigidity and the zero lower bound

Read:

- The figure compares the flexible-price economy with an economy with fixed nominal wages.
- Because the nominal rate cannot go negative enough, the zero lower bound prevents the real rate from falling to the flexible-price natural rate.
- Output therefore falls more — about 1.7% instead of 1.1% — and the recovery in interest rates is delayed.
- The slow deleveraging under the ZLB is not “good” deleveraging; it reflects depressed income rather than a weaker balance-sheet adjustment.

5.6 Figure IX through Figure XII: Alternative calibrations

Read:

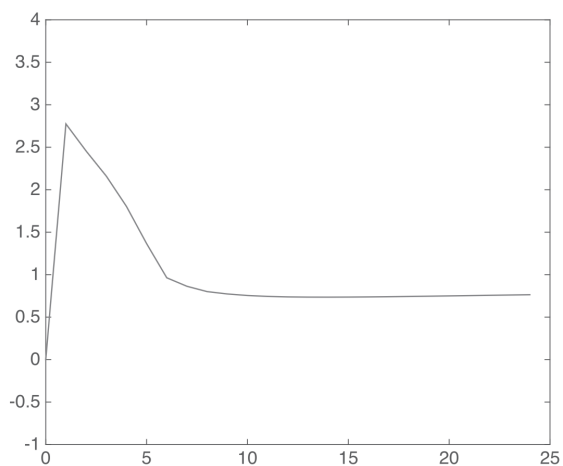


FIGURE VII
Employment Response
Percent deviations from initial steady state.

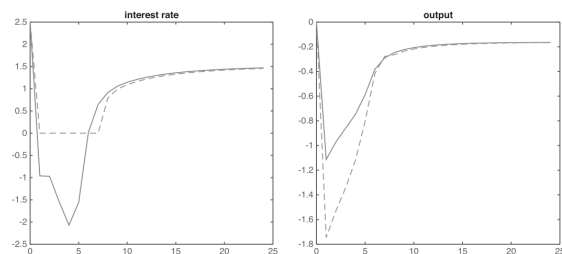


FIGURE VIII
Responses with Fixed Wages
Solid line: flexible price economy. Dashed line: economy with fixed wages.
Interest rate in annual terms. Output in percent deviation from initial steady state.

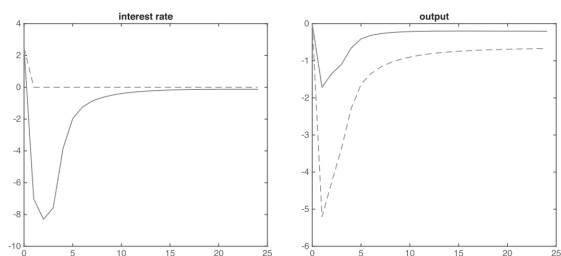


FIGURE IX
Interest Rate and Output: Median Wealth Calibration
Solid line: flexible price economy. Dashed line: economy with fixed wages.
Interest rate in annual terms. Output in percent deviation from initial steady state.

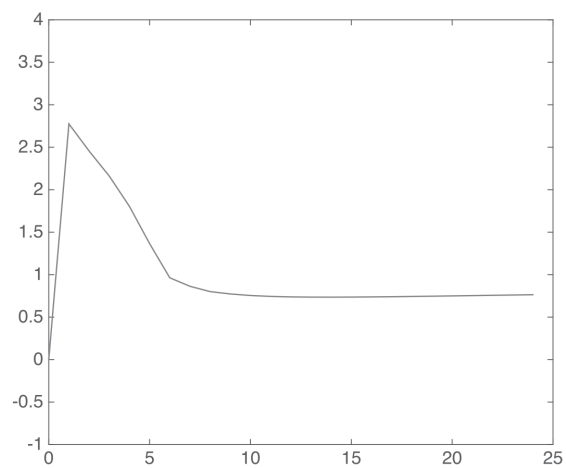


FIGURE VII
Employment Response
Percent deviations from initial steady state.

- Figure IX shows that targeting median liquid wealth makes the recession much deeper and the ZLB much more important.
- Figures X and XI show that changing labor-supply curvature can flip the employment response from positive to negative and can even generate a permanently negative natural rate.
- Figure XII shows that higher risk aversion deepens the downturn by weakening intertemporal substitution and making the interest-rate decline more necessary.
- Taken together, these figures say that the paper’s mechanism is robust, but its quantitative strength depends a lot on how rich or poor the “typical” household is in the calibration.

5.7 Figure XIII and Figure XIV: Fiscal policy and debt deflation

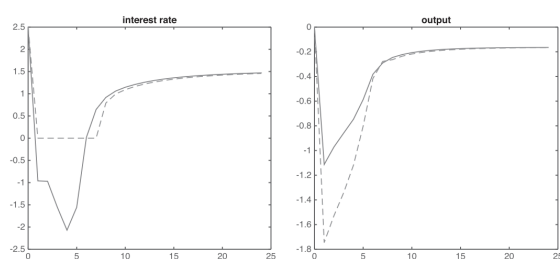


FIGURE VIII

Responses with Fixed Wages

Solid line: flexible price economy. Dashed line: economy with fixed wages. Interest rate in annual terms. Output in percent deviation from initial steady state.

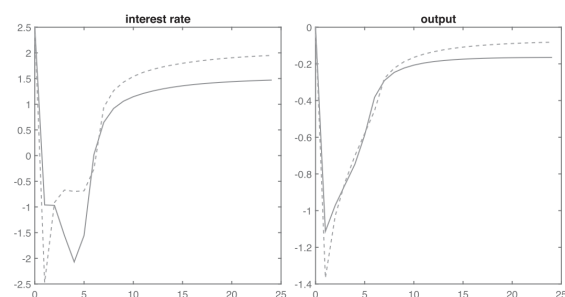


FIGURE XIV

Deflation: the Fisher Effect

Solid line: baseline. Dashed line: 10% deflation at $t = 0$. Interest rate is in annual terms. Output is in percent deviation from initial steady state.

Read:

- Figure XIII shows that debt-financed transfers cushion the downturn by supporting demand and expanding the supply of liquid assets.
- Transfers targeted to the unemployed are more effective than an equal-sized untargeted tax cut, because they reach the most constrained households.
- Figure XIV shows that unexpected deflation worsens the crisis by raising the real debt burden of borrowers.
- This is the paper’s Fisher debt-deflation lesson: increasing the real value of nominal debt in a deleveraging episode is contractionary.

5.8 Figure XV and Figure XVI: Durable goods and collateralized borrowing

Read:

- Figure XV shows that a permanent tightening in collateral constraints need not reduce total output if the fall in interest rates strongly boosts durable demand by net savers.
- In that experiment, nondurable spending falls but durable spending rises enough that output increases slightly.

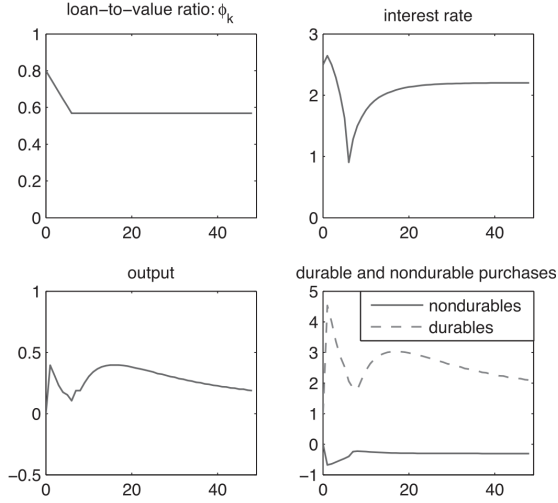


FIGURE XV

Responses to a Shock to the Borrowing Limit ϕ_k

Interest rate is in annual terms. Quantities are in percent deviation from initial steady state.

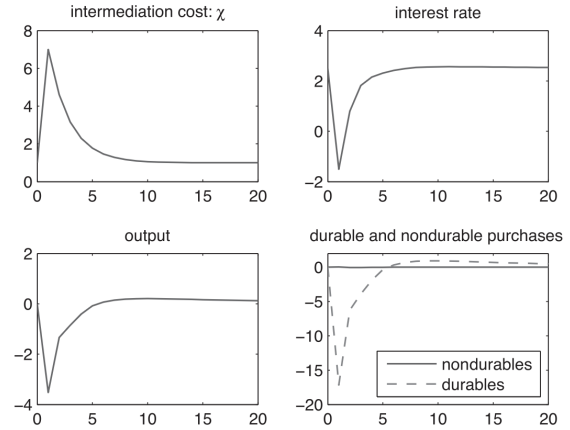


FIGURE XVI

Responses to a Temporary Shock to the Intermediation Cost χ

Interest rate is in annual terms. Quantities are in percent deviation from initial steady state.

- Figure XVI shows a very different shock: a temporary increase in borrowing spreads causes a large, short-lived collapse in durables and a much deeper contraction.
- The comparison of the two figures is the main lesson of the durables extension: what matters is not only the size of the credit shock but also whether it hits a narrow set of constrained borrowers or all borrowers through spreads.

6 Short summary

- **Conclusion.** A credit crunch in a heterogeneous-agent economy generates both deleveraging by borrowers and higher precautionary saving by lenders, which pushes the natural interest rate down sharply.
- **Main mechanism.** The full wealth distribution is essential. It creates interest-rate overshooting and internal persistence because many low-wealth households try to save simultaneously after the shock.
- **Main macro lesson.** Even with flexible prices, output can fall because lower interest rates only partially offset the demand contraction. With nominal rigidity and the zero lower bound, the contraction is larger.
- **Policy lesson.** Targeted fiscal support is more effective than untargeted transfers, while debt deflation worsens the downturn.
- **Extension lesson.** In a world with liquid and illiquid assets, credit shocks can move different components of expenditure very differently, especially durable spending.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- A tightening of household borrowing capacity can by itself produce a large fall in the natural interest rate.
- The reason is not only that constrained borrowers reduce debt, but also that unconstrained households raise precautionary savings.
- This mechanism can generate a recession even in a flexible-price economy, and it can push the economy into a liquidity trap once nominal rigidities are introduced.
- The dynamics are persistent because the wealth distribution adjusts slowly over time.

7.2 Economic intuition

The paper's deepest intuition is simple: a credit crunch does not just hurt borrowers. It changes the desired portfolios of the entire household sector. Poor households want to move farther from the borrowing limit. Rich households also want more buffer assets because future shocks are harder to smooth. Since the supply of safe liquid assets is fixed, the only way to make everyone willing to hold the existing stock is for the interest rate to fall. It falls especially sharply at first because the economy begins with a wealth distribution that is too concentrated near the lower tail relative to the new steady state. That is why the natural rate overshoots downward.

On the goods side, the lower interest rate is stabilizing, not destabilizing: it offsets part of the consumption collapse. The recession occurs because this offset is incomplete. Once nominal rigidities prevent the interest rate from falling enough, the same balance-sheet shock becomes much more contractionary.

7.3 Takeaway

This paper is best read as a heterogeneous-agent balance-sheet theory of credit-crisis recessions. Its main contribution is to show that credit-market disruptions naturally generate the type of negative natural-rate shock that liquidity-trap models need. But unlike representative-agent models that impose this force through preference shocks, Guerrieri and Lorenzoni derive it from household heterogeneity, borrowing constraints, and precautionary behavior. For work on deleveraging, household balance sheets, the zero lower bound, or secular stagnation, this paper is a benchmark reference.